

Energy & Sustainability Dashboard

Decarbonization and Renewable Adoption (1990-2024) - BI Portfolio Report

Focus: How emissions intensity, renewable uptake, and economic development interact across major economies.

Audience: ESG leaders, strategy teams, analysts, and policy stakeholders.

Primary deliverable: Interactive dashboard (browser-based) + this narrative report.

Last updated: 2025-12-31.

Why this topic matters: In 2025, sustainability analytics is a top BI theme because the energy transition affects cost, risk, regulation, supply chains, and capital allocation. Decision-makers need clear KPIs and a consistent story - not fragmented charts.

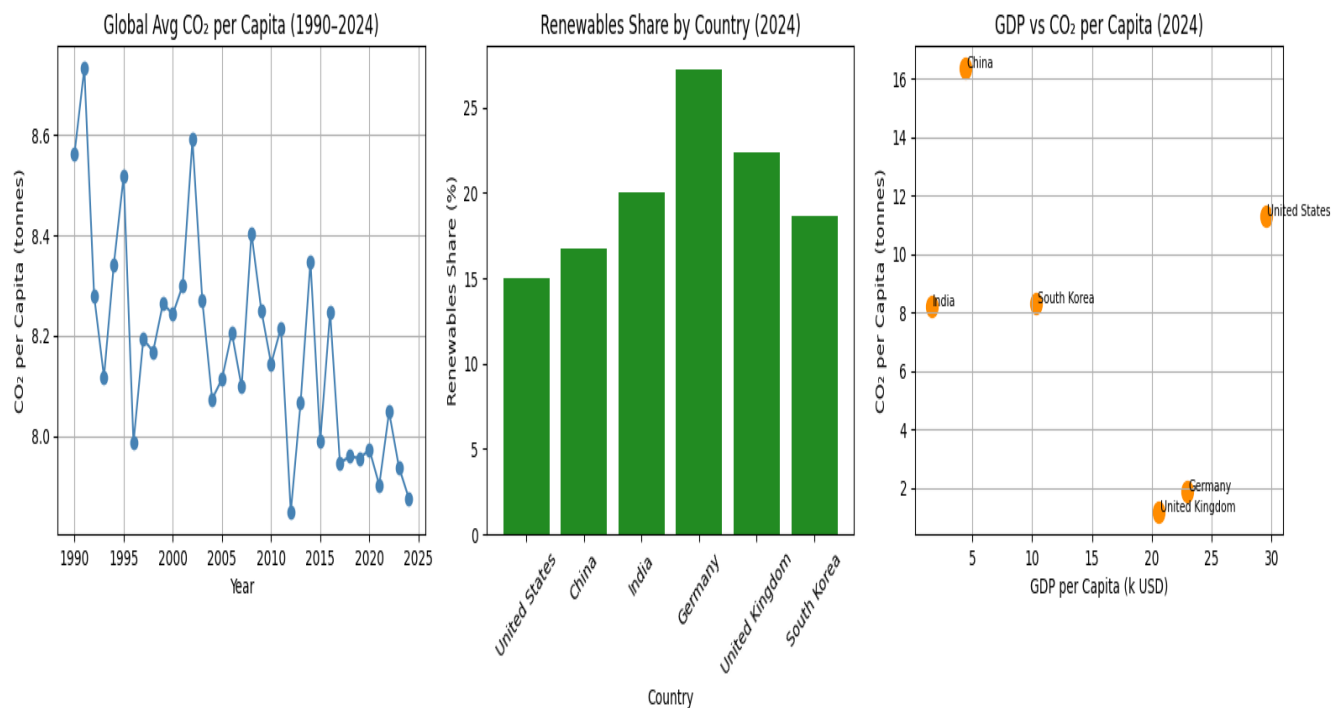


Figure 1. Dashboard snapshot (static export).

Executive Summary

This portfolio project demonstrates end-to-end BI delivery: problem framing, metric definitions, visual design, interactive exploration, and written storytelling. The dashboard consolidates three high-signal indicators - CO2 emissions per capita, renewable energy share, and GDP per capita - to help users answer a practical question: *Are countries achieving economic growth while reducing carbon intensity?*

What you can do with the dashboard: compare trajectories over time (1990-2024), filter to a single country using a dropdown slicer, and interpret the latest-year positioning in a GDP-vs-CO2 plot where bubble size represents renewables.

- The sample demonstrates mixed transition profiles: some economies reduce per-capita emissions as renewables rise (partial decoupling), while others still show high emissions intensity.
- Renewable adoption trends differ materially across countries, suggesting that policy, grid readiness, and industrial structure strongly shape outcomes.
- A single latest-year scatter view makes trade-offs obvious: the most transition-ready position is high income, low emissions, and high renewables (large bubble).
- The narrative layer (this report) explains how to interpret patterns and convert them into decisions (targets, investments, risk monitoring).

1. Business Context, Stakeholders, and Use Cases

Sustainability has moved from a compliance checkbox to a core driver of enterprise strategy. Transition risk (policy, technology, and market shifts) and physical risk (climate impacts) are increasingly priced into capital allocation. BI adds value by turning complex sustainability concepts into consistent, auditable metrics that leadership can act on.

Stakeholders

- **ESG / Sustainability Manager:** monitors progress against targets, prepares disclosures, and needs consistent KPI definitions.
- **Strategy & Finance:** evaluates transition investments and compares peer trajectories for prioritization.
- **Operations / Supply Chain:** uses macro indicators to contextualize decarbonization plans and supplier engagement.
- **Policy & Public Affairs:** benchmarks national progress to anticipate regulation and incentives.

What decisions this dashboard supports

- **Benchmarking:** identify leaders (high renewables, low CO2) and laggards for peer comparisons.
- **Target setting:** set realistic annual improvement goals based on leader trajectories.
- **Risk monitoring:** flag high-CO2/low-renewables profiles as potential transition-risk hotspots.
- **Storytelling:** translate complex trends into executive-ready narratives for reviews and reporting.

2. Data Model and KPI Definitions

The underlying dataset is a tidy fact table with a yearly grain. In a Power BI implementation, this table would connect to a Calendar dimension and a Country dimension (star schema). The three measures are defined with units and interpretation guidance below.

Field	Unit	Meaning (how to interpret)
CO2 per capita	t/person	Emissions intensity. Lower is better (holding context constant). Best used for trend + peer comparisons; does not capture total emissions.
Renewables share	%	Energy transition momentum. Higher indicates greater reliance on renewables; compare growth rate over time and cross-country gaps.
GDP per capita	k USD	Economic development proxy. Used to contextualize whether emissions reduction occurs alongside growth (decoupling signal).

Analytical assumptions

- Values are synthetic and calibrated for plausible directionality (not intended as factual measurements).
- Yearly grain supports long-term trend reading and avoids overfitting noise.
- The dashboard focuses on intensity metrics (per capita) to compare economies of different sizes.

3. Dashboard Guide (Functional Walkthrough)

The dashboard is intentionally simple: one slicer and three visuals. This reduces cognitive load and makes the analysis easy to explain in interviews. The dropdown behaves like a Power BI slicer: it toggles between a multi-country view and a single-country focus.

How to interact

- 1 Start with **All Countries** to compare long-term trajectories and identify outliers.
- 2 Select one country to isolate its emissions and renewables trend without visual clutter.
- 3 Use hover tooltips to read exact values at any year or point.
- 4 Return to All Countries to validate whether the focused story generalizes.

Visual 1 - CO2 per capita (line)

What it answers: *Is emissions intensity going down?* Use slope direction and turning points to discuss policy and structural changes. For BI storytelling, call out periods of plateau vs improvement.

Visual 2 - Renewables share (dotted line)

What it answers: *Is the energy mix shifting?* Pair this with CO2 trends to discuss whether renewables growth coincides with emissions reductions or whether demand growth offsets progress.

Visual 3 - GDP vs CO2 (latest year)

What it answers: *Is the country in a transition-ready position?* Interpret quadrants: high GDP + low CO2 suggests decoupling; low GDP + high CO2 suggests carbon-intensive growth. Bubble size adds a third signal for renewables.

Pro tip for interviews: describe a short analysis loop: (1) compare countries, (2) filter to one for a story, (3) return to confirm benchmark gaps, (4) translate into actions and next metrics.

4. Insight Narrative (What to Say When Presenting)

Below are example narrative statements that align with the visuals. The goal is to demonstrate BI communication: state the pattern, explain why it matters, and propose what to do next.

Long-term average emissions

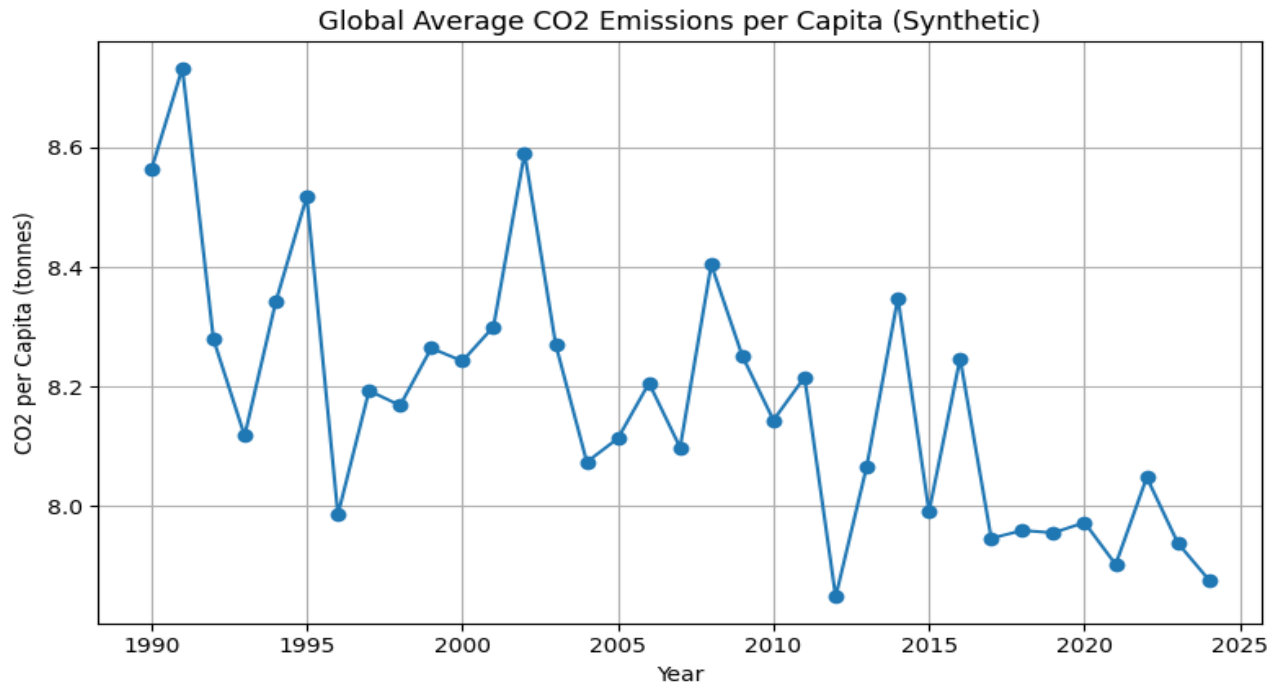


Figure 2. Global average CO2 emissions per capita (synthetic, 6-country mean).

Renewables adoption (latest year)

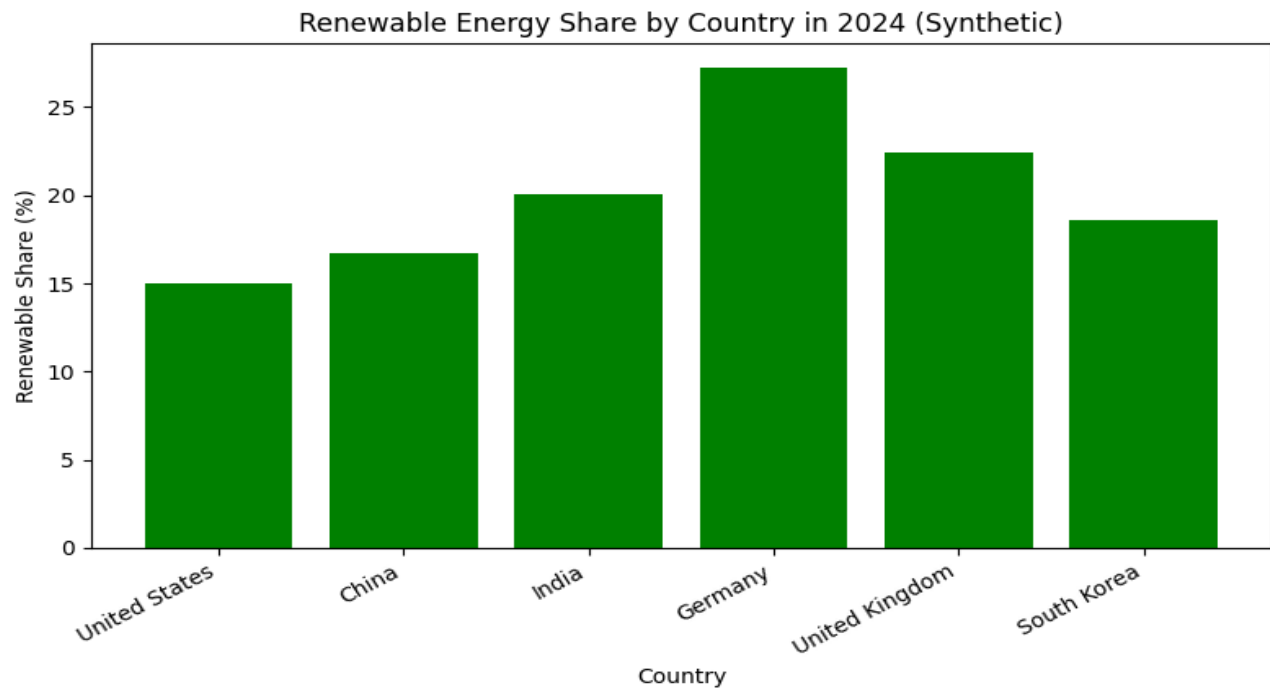


Figure 3. Renewable share by country in 2024 (synthetic).

Decoupling snapshot

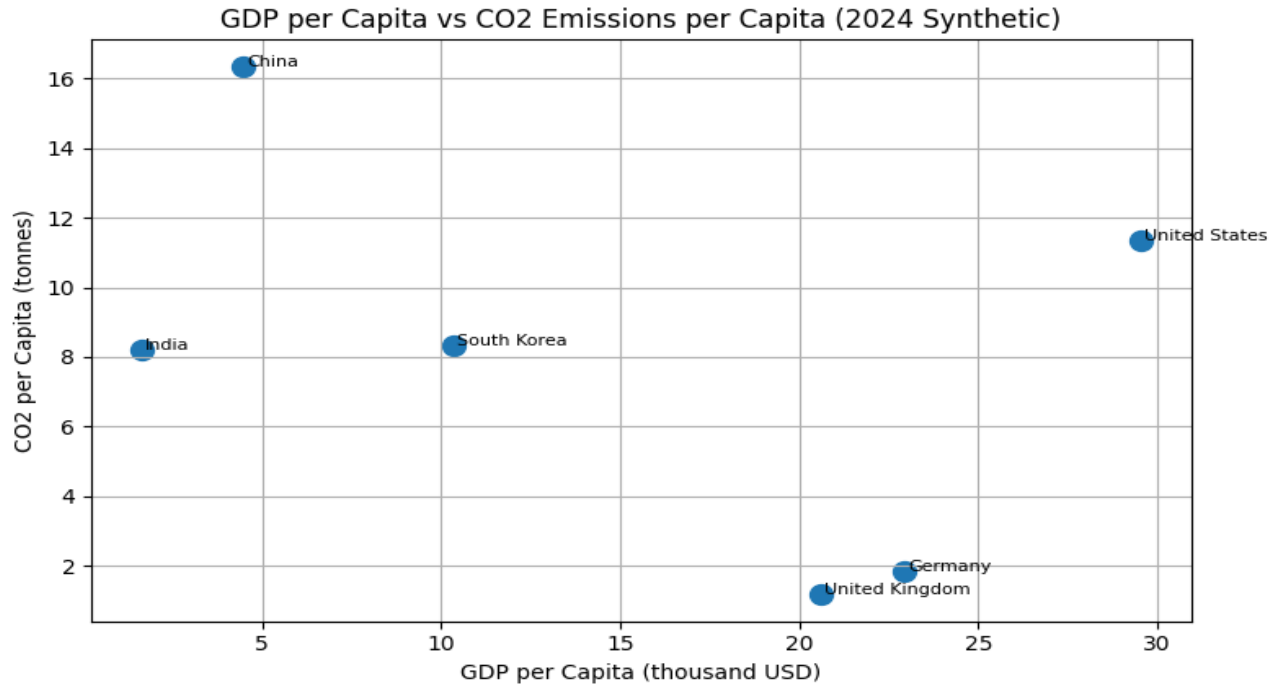


Figure 4. GDP vs CO2 per capita in 2024; bubble size approximates renewables share.

Example insight statements

- **Trend signal:** The multi-decade view suggests that some economies steadily reduce emissions intensity while others increase - which means progress is not uniform and should not be summarized by a single global average.
- **Renewables gap:** Countries show materially different renewable shares in the latest year. In practice, this often reflects policy design, market structure, and grid constraints; leaders can be used as benchmark targets for laggards.
- **Decoupling:** The GDP-vs-CO2 scatter makes it easy to explain that economic prosperity does not require high carbon intensity. The preferred region is high GDP, low CO2, plus strong renewables (larger bubble).
- **Actionable next metric:** Add total emissions and electricity mix to separate intensity improvements from scale effects (population and demand growth).

5. 2024 KPI Snapshot

This table mirrors what is visible via tooltips in the scatter plot and provides a quick executive summary for the latest year.

Country	CO2 per Capita (t)	Renewables Share (%)	GDP per Capita (k USD)
United States	11.3	15.0	29.5
China	16.4	16.7	4.5
India	8.2	20.1	1.6
Germany	1.9	27.2	22.9
United Kingdom	1.2	22.4	20.6
South Korea	8.3	18.6	10.4

What this table is good for

- Executive KPI readout (quarterly/annual review).
- Peer ranking (leaders vs laggards) before deeper drill-down.
- Baseline for setting forward-looking targets and scenario assumptions.

6. Recommendations (What to Build Next)

To make this dashboard portfolio-ready for real-world use, the next iteration should focus on stronger decision support: more KPIs, richer drill-down, and clearer governance.

Product improvements

- **KPI cards:** latest CO2, renewables, YoY delta, and 5-year change.
- **Drill-down:** add sector breakdown (power, transport, industry) and electricity mix.
- **Targets vs actual:** show progress toward a 2030/2050 pathway with variance metrics.
- **Annotations:** overlay policy milestones (carbon price, renewable subsidies) on time series.

Data engineering and governance

- Connect to authoritative sources (World Bank, Our World in Data) with scheduled refresh.
- Track data lineage: source, extract date, transformations, and QA checks.
- Document metric definitions to prevent inconsistent reporting across teams.

7. Limitations and Portfolio Notes

This report uses synthetic data and is intended as a demonstration of BI craft: modeling, visualization, interactivity, and storytelling. The same design can be reused with real datasets by keeping the schema stable and swapping the data source.

What I would say in an interview

- I designed the dashboard to answer one clear question (decoupling) with three supporting KPIs.
 - I prioritized interpretability: consistent colors per country and visual hierarchy (solid vs dotted).
 - I can extend this to real data and productionize refresh, QA, and governance.
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References (context sources used in the earlier draft)

- World Economic Forum (2025): energy transition progress in charts.
- Council Fire (2025): sustainability recap and lessons for business.
- KPMG (2025): supply chain trends (analytics + GenAI).